

ChronoStock

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Abstract

Retail investors often struggle to accurately interpret historical stock price movements. This is because existing financial applications tend to present quantitative price data in isolation, without providing the necessary context to fully understand it. As a result, individuals are left to manually aggregate and analyze extensive volumes of historical news and public discourse, a process that is fundamentally impractical for the average user. This report outlines the development of ChronoStock, a market intelligence platform that uses artificial intelligence to help visually contextualize past market fluctuations. Specifically, the system automatically extracts, summarizes, and ranks publicly accessible information to produce a structured, chronological account of the catalysts that shaped past price movements. Taken together, ChronoStock seeks to diminish the existing barriers to informed financial decision-making, equipping users with an intuitive mechanism to investigate the complex interplay between exogenous events, market sentiment, and historical stock performance.

1 Introduction and Problem Statement

When retail investors utilize standard financial applications to review the historical price charts of unfamiliar equities, they consistently face difficulties in discerning the specific events or qualitative factors that precipitated past price volatility. Comprehending an asset’s behavioral patterns over an extended temporal horizon necessitates the interpretation of a vast corpus of background information. However, the current landscape of financial technology exhibits critical deficiencies in facilitating this analytical process. Existing market applications predominantly display quantitative price charts entirely divorced from their qualitative context, requiring users to independently source and synthesize relevant background data.

Consequently, to understand long-term stock behavior, individuals must execute exhaustive manual searches across highly fragmented sources of historical news, corporate announcements, and market data. Even in instances where such extensive information is theoretically accessible, retail investors lack the sophisticated filtering mechanisms required to isolate the data that demonstrates genuine relevance to a specific asset’s valuation. Furthermore, contemporary platforms fail to offer systemic support for aligning this filtered qualitative data with corresponding quantitative price movements over time. Because users cannot readily observe the correlation between external market events and historical performance, delineating the causal factors behind specific price depreciations or appreciations becomes highly problematic. This distinct lack of an integrated, systematic methodology establishes a substantial barrier for retail investors attempting to extract actionable insights from historical market data, thereby degrading the user experience and impeding informed, empirical decision-making.

2 Related Work

2.1 Commercial Financial Platforms and Industrial Applications

The contemporary landscape of financial technology is characterized by platforms that exhibit a pronounced recency bias, prioritizing immediate market catalysts at the expense of comprehensive historical analysis. For instance, ubiquitous retail tracking platforms such as Yahoo Finance [6] actively employ artificial intelligence to generate real-time, aggregated summaries of concurrent news headlines. As observed in current iterations of the interface, automated modules like Yahoo Scout furnish users with immediate, highly condensed synopses of active corporate challenges, such as ongoing class action litigation or impending product announcements, coupled with analyst sentiment and revenue data.

However, this application of summarization technology is almost exclusively concentrated on current events, structurally neglecting the retrospective contextualization necessary for investors to evaluate long-term market trends. Within these architectures, qualitative information is sequenced chronologically by the most recent publication, a design paradigm that severely restricts user access to the historical events that previously dictated price volatility. Similarly, market visualization tools like Finviz [3] employ an aggregated methodology, presenting investors with an overwhelming volume of raw headlines devoid of sophisticated temporal synthesis.

Professional-grade analytics platforms, including Koyfin [4], offer extensive historical data and advanced charting capabilities but fundamentally lack the algorithmic capacity to directly integrate the semantic impact of historical press releases within the temporal axes of price charts. Furthermore, specialized applications such as Stock Events [5] focus exclusively on prospective market catalysts, including impending dividends, initial public offerings, and earnings reports, thereby omitting the retrospective historical context necessary to empirically analyze past volatility. Ultimately, these industrial solutions require the user to manually correlate fragmented historical data streams, underscoring the critical need for a unified, narrative-driven visualization approach that elevates past market events to the same level of accessibility as current news.

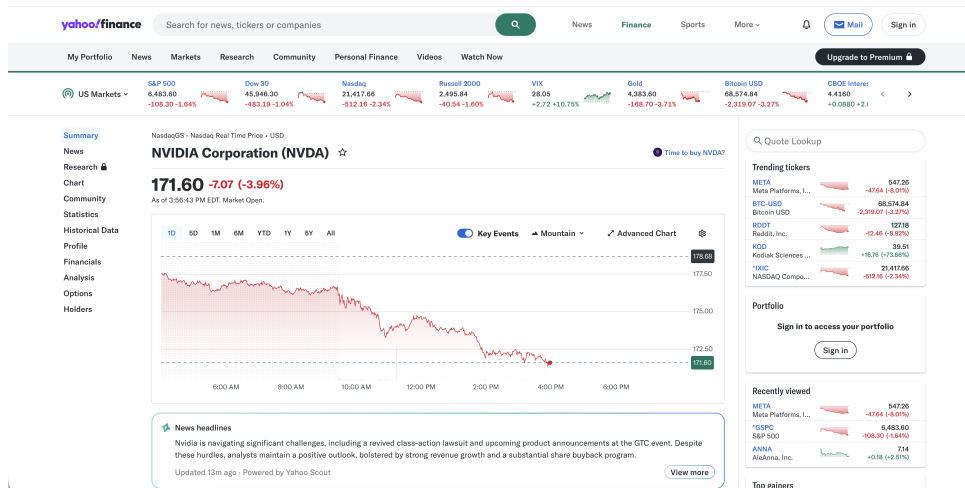


Figure 1: Screenshot of the Yahoo Finance interface.

2.2 Academic Methodologies and Research

Within the academic domain, a substantial corpus of literature investigates the complex relationship between financial news and equity valuation. Historically, scholars have predominantly concentrated on applying advanced Natural Language Processing architectures to perform pure sentiment analysis on textual news data. Models such as FinBERT [1] are explicitly trained to

extract and classify directional sentiment vectors from financial publications. However, a critical limitation of these methodologies is their tendency to evaluate news sentiment in strict isolation, completely divorced from the contextual reality of concurrent historical price ranges. By treating textual sentiment as an independent variable without anchoring it to localized quantitative performance, these studies often fail to capture the nuanced dynamics between isolated market events and actual price execution over time.

One line of research attempts to address this limitation by incorporating market price signals into the analysis of financial news. Del Corro and Hoffart propose a weakly supervised framework in which a neural network is trained to predict stock price movements using sequences of news headlines together with historical price information, and the internal attention mechanism is subsequently repurposed to estimate the financial relevance of individual news items [2]. In this formulation, attention weights learned during the prediction task are interpreted as indicators of how strongly a particular headline contributes to the model’s decision, enabling the ranking of news articles according to their inferred importance. Despite integrating price information, the methodology presents several limitations. The approach depends heavily on the performance and stability of the underlying prediction model, and attention weights do not necessarily provide a reliable or causal explanation of financial relevance. As a result, the inferred relevance scores may reflect correlations learned during training rather than genuine market-moving events. Furthermore, the method primarily operates on short textual inputs such as headlines and does not explicitly model discrete market events or structural shifts in price behavior over time.

3 Methodology

3.1 Overview

The objective is to identify news articles that correspond to core historical events of a company by combining financial market signals with semantic analysis of news content. The assumption is that important corporate events are reflected in abnormal stock returns or volatility, and that factual news published around these movements can explain the cause. The pipeline consists of five stages: (1) data collection, (2) keyword-based news filtering, (3) change point detection, (4) event window construction and scoring, and (5) LLM-based event validation.

3.2 Data Collection

The system uses two data sources: daily stock prices (from which returns, absolute returns, and volatility are computed) and company-related news articles with publication dates and headlines. The former identifies abnormal market movements; the latter provides candidate explanations.

3.3 Keyword-Based News Filtering

To reduce noise prior to event alignment, we apply a rule-based filter. For each company s , we define a keyword dictionary $\mathcal{K}_s$ containing the company name, ticker, major products, subsidiaries, and key executives. An article is retained only if its title or content contains at least one keyword from $\mathcal{K}_s$. An exclusion filter further removes opinion pieces, analyst commentary, and speculative content unlikely to describe concrete corporate events.

3.4 Change Point Detection in Market Signals

We detect structural changes in market behavior using the Pruned Exact Linear Time (PELT) algorithm, applied independently to two derived signals. Let P_t denote the closing price at time

t . The daily log-return is $r_t = \log(P_t/P_{t-1})$. The first signal is the normalized absolute return:

$$\tilde{a}_t = \frac{|r_t| - \mu_a}{\sigma_a},$$

which captures large price moves regardless of direction. The second is rolling volatility:

$$v_t = \sqrt{\frac{1}{w} \sum_{i=t-w+1}^t (r_i - \bar{r}_t)^2},$$

which captures periods of sustained uncertainty. Together, these signals detect both sharp single-day reactions and gradual regime shifts.

PELT finds the optimal set of change points $\tau = \{\tau_1, \dots, \tau_m\}$ by minimizing:

$$\min_{\tau} \left(\sum_{i=0}^m \mathcal{C}(y_{(\tau_i+1):\tau_{i+1}}) + \beta m \right),$$

where $\mathcal{C}(\cdot)$ is a segment cost function and β is a penalty controlling sensitivity. Change points are detected separately for each signal, and the final candidate set is their union $C = C_r \cup C_v$, improving robustness by capturing both return-driven and volatility-driven structural changes.

3.5 Event Window Construction and Scoring

For each change point c_i , an event window $W_i = [c_i - L, c_i + R]$ is constructed to capture surrounding news. Overlapping windows are merged to avoid fragmenting clusters of related market reactions. Each window is scored by the absolute cumulative abnormal return:

$$|CAR_{t_1, t_2}| = \left| \sum_{t=t_1}^{t_2} (R_t - E[R_t]) \right|,$$

where $E[R_t]$ is the expected return from a market benchmark model. Larger $|CAR|$ indicates stronger market responses, prioritizing windows more likely to correspond to significant real-world events.

3.6 LLM-Based News Relevance Filtering

Candidate news articles within each event window are evaluated by a large language model. Unlike keyword or statistical methods, the LLM analyzes the semantic relationship between the article content and the observed price movement. Specifically, it assesses whether the article (a) is factual rather than opinion-based, (b) describes a concrete company-level event, and (c) plausibly explains the observed market reaction. Opinion pieces, speculative articles, and redundant reports are discarded, producing a final set of validated event explanations.

4 System Design

4.1 Overview

ChronoStock is implemented as a hybrid **online serving system** and **offline data pipeline**. The user-facing path is a web application built with Next.js that communicates with a FastAPI backend. The backend serves stock charts, news, macro indicators, authentication, and watchlist operations. For frequently updated data such as prices and trending tickers, the backend fetches information from external providers and caches results to reduce latency.

High-value event intelligence is generated by a background pipeline that periodically ingests raw market data, detects candidate price-moving events, filters them using a large language model (**DeepSeek-Chat**), and stores the curated events for fast retrieval. This separation ensures the backend remains lightweight while providing precomputed, explainable event overlays to users.

4.2 Serving Architecture

At runtime, the frontend communicates solely with the backend API. The backend exposes endpoints for prices, ticker search, stock metadata, raw news, SEC filings, macro indicators, market analysis, and user operations. Curated event overlays are served from the `stock_events` table for direct rendering on the stock timeline. Offline pipeline tasks perform the computationally heavy processing, keeping online serving low-latency.

4.3 Medallion Data Architecture

We adopt a **medallion architecture** to organize the pipeline stages:

- **Bronze layer (S3 standard):** stores raw stock prices and news exactly as collected.
- **Silver layer (S3 standard):** stores cleaned and filtered news.
- **Gold layer (RDS PostgreSQL db.t3.micro):** stores LLM-filtered event records; the relational database provides indexed lookups for fast backend queries.

This separation enables reproducibility, auditability, and recoverability. Changes in LLM prompts or heuristics can be applied without re-ingesting raw data.

4.4 Compute and Deployment

The system uses a single EC2 instance (`t2.micro`) running both the backend Docker container and scheduled pipeline jobs. This choice balances simplicity and cost for the scale of the project, ensuring consistent environments and minimal operational overhead. Bootstrapping scripts configure Docker, the SSM agent, CloudWatch agent, and environment variables.

4.5 Scheduling and Background Execution

Pipeline tasks are scheduled via **Amazon EventBridge Scheduler** and **AWS Systems Manager**:

- Daily pipeline for news and updates
- Hourly price refresh
- Monthly recomputation of events

EventBridge triggers commands on the EC2 instance, which executes the corresponding scripts. This approach avoids SSH-based automation and keeps execution close to the application environment and storage.

4.6 Security and Observability

- EC2 runs in a public subnet for frontend traffic; PostgreSQL RDS runs in private subnets, accessible only from the EC2 instance.
- S3 buckets block public access; IAM policies restrict operations to the configured pipeline.
- Secrets (database credentials, API keys) are managed in AWS Secrets Manager.
- CloudWatch collects logs for backend and pipeline jobs, and alarms monitor CPU usage.

4.7 Design Rationale for Infrastructure Types

- **S3 standard:** provides durable, low-cost storage suitable for raw and intermediate datasets with append-heavy operations.
- **RDS db.t3.micro with PostgreSQL:** suitable for lightweight relational workloads with small storage needs; supports indexing and query efficiency for serving events.
- **EC2 t2.micro:** sufficient for the scale of the application, lightweight background jobs, and low-traffic serving; simplifies deployment and reduces cost.

4.8 Design Tradeoffs

This architecture emphasizes maintainability, reproducibility, and low-latency serving. Tradeoffs include a single EC2 instance (potential compute bottleneck), CSV-based S3 storage (less efficient than columnar formats), and offline LLM filtering (event explanations are precomputed rather than real-time). These tradeoffs are acceptable for the project’s goals while leaving room for future scaling.

5 Experiment and Results

5.1 Experimental Setup

This section describes how we evaluate the proposed pipeline for identifying important company events from financial news. Our experiments are designed to assess three aspects of the system: (1) how the pipeline filters and reduces large volumes of raw news data, (2) how accurately it identifies major corporate events compared to baseline methods, and (3) how individual components contribute to overall performance.

We conduct experiments using financial news and market data collected between April 2021 and March 2026 for seven major technology companies: AAPL, AMZN, GOOGL, META, MSFT, NVDA, and TSLA. The full pipeline is applied to the collected news corpus, including keyword-based filtering, change-point-based event window detection, CAR-based ranking, and LLM-based event validation.

To quantitatively evaluate event detection accuracy, we construct a ground-truth dataset by manually annotating major corporate events for the seven representative companies mentioned above. For each company, we identify 20-30 significant events from the raw news dataset. This results in a total of 195 annotated events spanning earnings announcements, product launches, mergers and acquisitions, regulatory developments, and executive changes. We evaluate performance using ranking-based metrics: **Precision**, **Recall**, and **F1**.

Using this setup, we perform several experiments. First, we analyze how each stage of the pipeline reduces the number of candidate news articles. Next, we conduct an ablation study to understand the contribution of each component in the pipeline. Finally, we present a qualitative case study on NVDA to illustrate how the system selects meaningful news events in practice.

5.2 Pipeline Performance Summary

To evaluate the effectiveness of each stage in the proposed pipeline, we measure how the number of news articles decreases as filtering and event selection are applied. Experiments were conducted on seven representative companies: AAPL, AMZN, GOOGL, META, MSFT, NVDA, and TSLA using news data collected between April 2021 and March 2026. Table 1 reports the average number of articles remaining after each processing stage.

Pipeline Stage	Average Articles	Remaining (%)
Raw collected news	18,882	100%
After keyword filtering	5,825	30.8%
Within detected event windows	407	2.2%
After LLM event filtering	20	0.1%

Table 1: Average pipeline reduction statistics across the "Magnificent Seven" stocks.

The results highlight the effectiveness of the multi-stage filtering pipeline. Keyword-based filtering significantly reduces irrelevant articles while preserving company-specific news. Change point detection further focuses the dataset on periods of abnormal market activity, reducing the candidate articles to a small fraction of the original corpus. Finally, LLM-based event validation selects only a limited set of high-quality articles that correspond to major company events. Overall, the system reduces the original dataset by more than 99% while retaining the most informative event-related news.

5.3 Change Point Visualization

Figure 2 illustrates the historical price of NVIDIA (blue curve) alongside the Dow Jones Industrial Average (red curve) as a market reference. The y-axis represents the stock price, and the vertical lines indicate detected change points, which serve as the centers of candidate event windows in our pipeline. Each window extends several days before and after the center, capturing the market response surrounding structural shifts.

Comparing NVIDIA’s price with the market index provides context for distinguishing stock-specific events from broader market movements. Periods where NVIDIA deviates significantly from the market trend correspond to large absolute returns or volatility spikes, which trigger change point detection. These detected points thus highlight intervals where meaningful company-specific events are likely to have occurred.

This visualization confirms that the PELT-based change point detection effectively identifies periods of abnormal stock behavior, forming the basis for subsequent news extraction and LLM-based event validation.

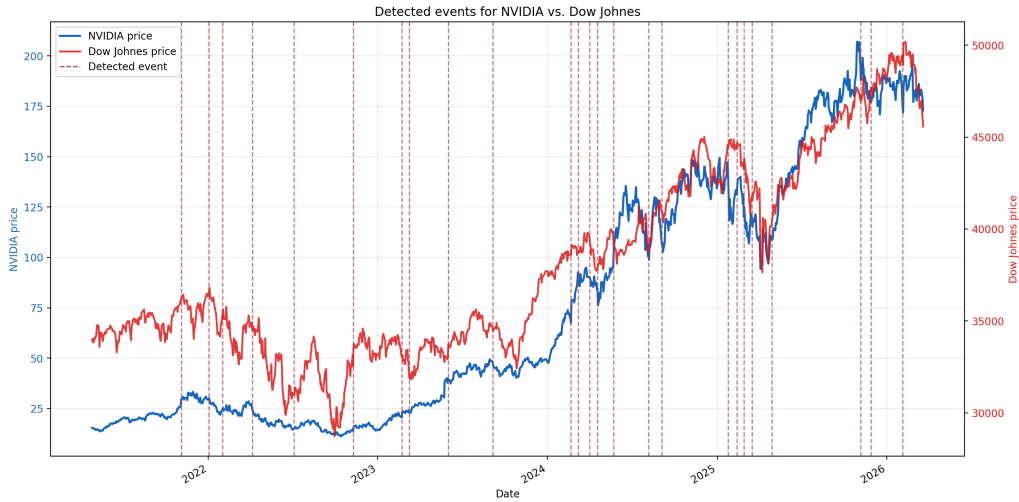


Figure 2: Historical stock price of NVIDIA (blue) versus Dow Jones Industrial Average (red). Vertical lines indicate the detected change points, which are the centers of candidate event windows.

5.4 Ablation Study

To understand the contribution of each pipeline component, we perform an ablation study in which one component is removed at a time while keeping all others fixed. Table 2 reports the results.

Configuration	Precision	Recall	F1
Full pipeline	0.1298	0.0872	0.1043
– LLM validation	0.0276	0.1412	0.0462
– CAR scoring	0.0563	0.0987	0.0717
– Volatility signal	0.1201	0.0732	0.0910
– Keyword filtering	0.0312	0.0851	0.0457
– PELT detection	0.0417	0.0522	0.0464

Table 2: Ablation study results averaged over five tickers. Each row removes one component from the full pipeline.

Removing LLM validation leads to the largest drop in precision, decreasing from 0.1298 to 0.0276, while recall increases to 0.1412. This behavior is expected because, without the final semantic filtering stage, a much larger number of articles are retained within each event window. As a result, the system captures more ground-truth events but also introduces a substantial amount of irrelevant or speculative news. These results indicate that the LLM validation stage plays a crucial role in improving the precision of the pipeline by filtering out articles that are not directly associated with meaningful corporate events.

Removing CAR scoring reduces the ability of the system to prioritize news associated with stronger market reactions. Precision decreases to 0.0563 and the overall F1 score drops to 0.0717. Without CAR-based ranking, articles within detected event windows are no longer ordered according to their estimated market impact, which causes less informative news to appear among the top-ranked results. This suggests that CAR scoring helps highlight events that are more closely aligned with abnormal market movements.

Removing the volatility signal slightly reduces both precision and recall, resulting in an F1 score of 0.0910. This indicates that while large return spikes remain detectable, the absence of volatility-based signals weakens the system’s ability to capture broader structural shifts in stock behavior. Some events that unfold over multiple days may therefore be missed when volatility information is excluded.

Removing keyword filtering significantly degrades performance, with precision dropping to 0.0312. Without the initial filtering step, the pipeline receives a large volume of unrelated or weakly related articles mentioning the company, which reduces the effectiveness of downstream components. This highlights the importance of early-stage noise reduction in large financial news corpora.

Removing PELT detection also results in substantial degradation, with both precision and recall falling to 0.0417 and 0.0522 respectively. Without change-point-based event window selection, the system loses its ability to focus on periods of abnormal market behavior, and candidate articles are effectively sampled from a much broader time range. This confirms that price-signal-based event detection is a key component of the pipeline.

Overall, the full pipeline achieves the best balance between precision and recall among all configurations, yielding the highest F1 score (0.1043). Although the absolute values of precision and recall appear relatively low, this is partly due to the nature of the evaluation setup. The ground-truth dataset is manually constructed and inherently subjective, as it is difficult to determine

which news articles represent the most significant indicators in a company’s historical timeline. In many cases, multiple news reports may describe the same underlying event, or important contextual articles may not be included in the annotated set.

Therefore, some extracted events that are relevant from a market or analytical perspective may not be counted as correct under the strict matching criteria used in the evaluation. As observed in the qualitative analysis, the events identified by the full pipeline align well with known major developments such as earnings announcements, partnerships, and product launches. This suggests that, despite the conservative quantitative metrics, the pipeline is effectively capturing meaningful event-driven news in practice.

5.5 Qualitative Analysis: NVDA News Events

Table 3: Head of NVDA news articles with qualitative significance labels

Published Date	Title	Significance	Type	Notes
2024-01-31	NVIDIA Sets Conference Call for Fourth-Quarter Financial Results	Medium	Calendar of Events / Conference Calls	CFO commentary
2024-02-22	10+ Stocks, ETFs To Ride On Nvidia’s Record-Breaking Success	High	News / Top Stories / Markets	AI earnings drive market impact
2024-03-07	NVIDIA and HP Supercharge Data Science and Generative AI on Workstations	Medium	Product / Services	Collaboration for AI workstation acceleration
2024-04-18	Nvidia’s New Artificial Intelligence (AI) Chip Could Be the Catalyst to Take the Stock to the Next Level	Medium	Investing	AI chip GB200 expected to boost revenue
2024-09-11	Nvidia CEO Jensen Huang addresses the big question on investors’ minds	Medium	Investor Commentary	CEO discusses AI revenue potential

To better understand the relevance of news selected by our pipeline for NVIDIA Corporation (NVDA), we present a subset of news items with publication dates, titles, significance levels, and news types. Notes provide brief context where needed. For brevity, only the head of the full table is shown here.

The NVDA news events encompass earnings announcements, product launches, strategic partnerships, regulatory developments, and investor commentary. High-significance items are generally associated with major financial results, multi-industry collaborations, or announcements likely to influence AI infrastructure adoption and market sentiment. Medium-significance items typically provide context, operational updates, or commentary that indirectly affects investor perceptions. This qualitative categorization helps identify which news stories are most likely to impact NVDA’s stock movement or strategic positioning.

The qualitative results indicate that our pipeline effectively captures a diverse set of relevant news events. High-significance events, such as major earnings announcements, strategic AI partnerships, and cross-industry collaborations, are consistently identified, demonstrating the pipeline’s capacity to highlight impactful developments. Medium-significance events, including conference calls, product updates, and investor commentary, are also captured, reflecting the system’s sensitivity to contextually important but less market-moving information.

Overall, the pipeline exhibits strong precision in selecting news aligned with known market-moving events. The presence of detailed notes linking events to broader market or operational impact suggests that the system considers not only headline content but also underlying relevance factors. Some medium-significance items, such as investor alerts or minor regulatory updates,

indicate that certain news may have limited immediate effect, reflecting a conservative strategy that favors coverage completeness over aggressive filtering.

In summary, the qualitative assessment confirms that the pipeline balances relevance and breadth: it identifies critical events while providing sufficient contextual information for analysts.

6 Alternative Approaches Explored

In addition to the final pipeline described earlier, we experimented with several alternative approaches for identifying and ranking relevant financial news. While these methods showed some promising aspects, they also introduced practical challenges that limited their effectiveness in our setting.

Sentence Embeddings with Neural Network Prediction. One alternative approach was to directly model the relationship between news content and market reactions using a neural network. Instead of selecting news based solely on the CAR score observed at the publishing date, which can be noisy and influenced by unrelated market movements, we attempted to learn a predictive model of price impact. Specifically, we generated sentence embeddings from the news title and description, and used these embeddings as input features to a neural network trained to predict the absolute CAR centered at the news publication time. The predicted CAR score was then used as a proxy measure of news importance, allowing us to rank news articles according to their estimated market impact.

While this approach provides a principled way to connect textual signals with market reactions, we observed a key limitation in practice. Financial news relevance is often highly ticker-specific: a semantic pattern that signals a strong price movement for one company may have little or no impact for another. As a result, the neural network struggled to generalize across different stocks, particularly when the training data contained heterogeneous event types and industry contexts. This reduced the stability of the rankings and made the model sensitive to the distribution of tickers present in the training data.

Word Embeddings with Engineered Features and LightGBM. We also experimented with a gradient boosting approach using LightGBM, combining word embeddings with manually engineered features derived from the news metadata and event alignment. The feature set included attributes such as news identifiers, associated tickers, publication time, proximity to detected events, CAR values, temporal distance to events, sentiment scores, and weak supervision labels used during earlier stages of the pipeline. The goal of this approach was to leverage both semantic signals and structured information to rank news items more effectively.

However, we observed that the resulting rankings tended to be highly clustered within specific time periods. In practice, the model frequently prioritized news items occurring close to certain detected events or market fluctuations, leading to a concentration of selected articles within narrow time windows. This behavior reduced the temporal diversity of the selected news and made it more difficult to obtain a balanced representation of important events over longer time horizons.

Overall, these experiments highlight the challenges of directly learning market relevance from text and metadata alone. While machine learning models can capture useful signals, issues related to generalization across tickers and temporal clustering suggest that additional structure or constraints are necessary for stable and interpretable event selection.

7 Future Work

While the current implementation of ChronoStock demonstrates the feasibility of integrating market signals with news-based event detection, several directions remain open for substantial improvement in future iterations.

7.1 Domain-Specific Model Training

In its present form, the system relies on general-purpose large language models such as Claude and DeepSeek for the semantic filtering and event validation stages of the pipeline. These models are accessed through external API calls, effectively positioning the current system as a wrapper around commercially available LLM inference services. Although this approach is sufficient for prototyping and proof-of-concept evaluation, it introduces several practical limitations, including dependency on third-party availability, latency constraints imposed by network round-trips, and the inability to fine-tune model behavior for the specific task of financial event extraction.

A critical direction for future work is the development and training of a domain-specific language model tailored explicitly to the task of identifying, classifying, and summarizing stock-relevant corporate events from unstructured news text. Such a model could be fine-tuned on curated datasets of financial news articles annotated with ground-truth event labels, enabling it to internalize domain-specific reasoning patterns that general-purpose models handle only approximately. Fine-tuning would also allow the model to better distinguish between factual event reporting and speculative commentary, a distinction that remains challenging for zero-shot prompting of general-purpose models. Furthermore, a locally deployed, task-specific model would eliminate API dependency, reduce inference costs, and enable tighter integration with the rest of the pipeline.

7.2 Higher-Quality Data Sources

The prototype is constrained to free-tier news and market data APIs, which suffer from rate limits, inconsistent metadata, temporal gaps, and duplicate articles. Integrating premium feeds from providers such as Bloomberg, Reuters, or Dow Jones would improve coverage and editorial quality, reducing noise before expensive downstream processing. Institutional-grade market data with proper corporate action adjustments (splits, dividends) would also improve change point detection reliability. Additionally, dedicated deduplication, entity resolution, and temporal normalization pipelines would ensure that surfaced events are both accurate and non-redundant.

7.3 Additional Directions

Beyond the two primary areas identified above, several other improvements warrant exploration. First, the system could incorporate alternative data sources such as social media sentiment, regulatory filings, and macroeconomic indicators to enrich the contextual information available for event interpretation. Second, the event scoring mechanism could be enhanced through the integration of learned ranking models that account for sector-specific dynamics and cross-asset correlations. Third, a user feedback loop could be introduced to allow investors to rate the relevance of surfaced events, providing supervised signal for iterative model improvement. Finally, expanding the platform to support real-time event detection, rather than purely retrospective analysis, would substantially broaden its applicability for active portfolio management and risk monitoring.

8 Reflection and Conclusion

Building ChronoStock gave our team the opportunity to apply nearly every major topic from this course in the context of a single, end-to-end ML system. Below we reflect on course connections, what we would change in hindsight, and key technical learnings.

8.1 Connection to Course Material

Infrastructure and Deployment Docker, Terraform, and cloud architecture lectures directly shaped our deployment. We containerized the backend, defined all AWS resources as Terraform modules, and scripted reproducible deployments. Real-world issues — networking between public and private subnets, certificate provisioning with Certbot, coordinating scheduled pipeline jobs on EC2 — taught us the operational complexity that lecture examples could only sketch. Separating the lightweight serving path from the heavy offline pipeline applied the scalability principles from Module 9.

Prompting and Constrained Decoding Our LLM-based filtering is fundamentally a prompt engineering problem. We iterated through familiar failure modes: hallucinated IDs, free-text instead of JSON lists, opinion articles slipping through. Explicit formatting instructions and negative examples, applying the constrained-decoding mindset, were essential to getting reliable, machine-parseable output.

Model Serving Consuming DeepSeek-Chat and AWS Bedrock as external APIs taught us client-side serving trade-offs: rate limit management, request batching, exponential backoff retries, and S3 result caching. The LoRA and quantization discussions motivated our future-work plan to fine-tune and self-host a domain-specific model.

Data Engineering and Feature Stores Our medallion architecture (bronze/silver/gold layers in S3 and PostgreSQL) mirrors course principles. Immutable raw storage let us re-run downstream stages freely when changing prompts or PELT parameters. Precomputed CAR scores and macro indicators function as materialized features with time-based expiry.

8.2 What We Would Change in Hindsight

Invest in evaluation infrastructure earlier. We spent the majority of development building the pipeline and frontend before establishing a systematic way to measure output quality. In retrospect, creating a small manually-labeled ground truth set at the start would have let us iterate on prompts and PELT parameters with quantitative feedback rather than relying on spot-checking. The lack of early evaluation infrastructure meant we could not confidently compare design alternatives until late in the project.

Decouple the LLM dependency sooner. We integrated the LLM as a synchronous call within the pipeline, which made the entire event generation process dependent on external API availability. When DeepSeek experienced rate-limit throttling during batch runs, the pipeline would stall for extended periods. A better design would have been to make the LLM step asynchronous from the start — writing candidate articles to a queue and processing them independently — so that upstream stages (data ingestion, change point detection) could proceed without blocking.

Start with fewer tickers and go deeper. We initially targeted 20 companies to demonstrate breadth, but this spread our data-quality efforts thin. Focusing on 3–5 tickers first would have let us build robust keyword dictionaries, validate change point sensitivity per sector, and establish per-company baselines before scaling horizontally.

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